

Predicting the structure of a ciliary GTPase

with the AlphaFold3 server

Computer exercise (Part 1): AI for structure prediction

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What you need (please prepare before the session)

A laptop with a web browser.

A Google account, already signed in. We use it for the free AlphaFold3 server (alphafoldserver.com); nothing to install, it runs in the browser.

PyMOL installed, and opened once to check it launches. Free educational build for Windows and Mac: pymol.org/edu.

Aim of this exercise

In this exercise you will make and interpret your own protein structure predictions using the **AlphaFold3 web server**, using the ciliary GTPase **ARL13B** as an example (a protein many of you will recognise as a widely used marker for the ciliary membrane). You will:

- submit two predictions: ARL13B alone (*apo*) and ARL13B with a bound **GTP** and **Mg²⁺** ligand (*holo*);
- read the **pLDDT** (per-residue confidence) and **pTM** (global fold) scores, in the browser and in **PyMOL**;
- use the five predicted models that AF output to see which parts of the protein are reliable (the core fold) and which are not (the flexible termini);
- probe how binding a ligand influences AlphaFold's confidence in the flexible **switch regions**;
- take a first look at the **PAE** plot as a measure of how well the relative positions of different parts of a model are known.

1 Background: ARL13B

ARL13B is a small GTPase of the ARF/ARL family that localises to the ciliary membrane; loss-of-function mutations cause the ciliopathy Joubert syndrome. Like all small GTPases it has a compact **G-domain** that binds guanine nucleotide (GDP/GTP) and a **Mg²⁺** ion, with two mobile **switch** loops that respond to the nucleotide. In addition:

- Two cysteines near the **N-terminus** (Cys8/Cys9) are palmitoylated, anchoring ARL13B to the ciliary membrane as a peripheral membrane protein. This stable membrane association is what makes it such a good marker for staining cilia.
- The long **C-terminal extension** is largely disordered and carries a ciliary-targeting motif.

The folded G-domain spans roughly residues 20–240; beyond about residue 250 the long C-terminal extension is largely disordered (about half of the whole protein).

2 Getting started

Task. Open a browser, go to <https://alphafoldserver.com> and sign in with a Google account. The server is free for non-commercial use.

3 Submit two predictions

Submit **both** jobs now, so they compute while we go through the theory.

Job 1, apo (protein only): start a new job, add one **protein** entity, paste the sequence below, name it ARL13B_apo, and run it.

HsARL13B sequence (paste this)

```
MFSLMASCCGWFKRWREPVRKVTLMLVGLDNAGKTATAKGIQGEYPEDVAPTGVGFSKINL
RQGKFEVTIFDLGGGIRIRGIWKNYAESYGVIFVVDSSDEERMEETKEAMSEMLRHPRI
GKPIVLVANKQDKEGALGEADVIECLSLEKLVNEHKCLCQIEPCSAISGYGKKIDKSIKKG
LYWLLHVIARDFDALNERIQKETTEQRALEEQEKQERAERVRLREERKQNEQEQAEIDGT
SGLAELDPEPTNPFQPIASVIIENEGKLEREKKNQKMEKDSGCHLKHKMEHEQIETQGGV
NHNGQKNNEFGLVENYKEALTQQLKNEDETDRPSLESANGKKTKKLRMKNHRVEPLNID
DCAPESPTPPPPPPVVGWGPVKVTRLPKLEPLGETHHNDFYRKPLPLAVPQRPNSDAHDVIS
```

Job 2, holo (protein + GTP + Mg²⁺): start another new job, add the same protein entity, then add a **ligand** entity and select **GTP**, and an **ion** entity and select **Mg²⁺**. Name it ARL13B_GTP and run it.

Note. Each job may take a few minutes to a while depending on server load. Time for a short introduction to AlphaFold3 while your jobs are running.

4 While it runs: what is AlphaFold doing?

AlphaFold does not simulate folding. It has learned, from the whole Protein Data Bank, to predict 3D coordinates directly from sequence and from an alignment of related sequences. Crucially it also predicts *how confident it is*:

- **pLDDT**: a *per-residue* score (0–100); the model is coloured by it. High = the local geometry is reliable.
- **pTM**: one *global* score (0–1) for the overall fold and domain arrangement.

5 First look in the browser: pLDDT and pTM

When ARL13B_apo finishes, open it. The 3D viewer colours the model by pLDDT, on the AlphaFold scale:

■ blue	very high (pLDDT > 90)
■ cyan	confident ($90 \geq \text{pLDDT} > 70$)
■ yellow	low ($70 \geq \text{pLDDT} > 50$)
■ orange	very low (pLDDT ≤ 50)

Question. Rotate the model and look at the pLDDT colouring: which parts are blue/cyan and which are yellow/orange? Then compare this with what you know about ARL13B from Section 1: do the confident and the low-confidence regions match the folded G-domain and the disordered C-terminal extension? The result also reports a single overall **pTM**: for a protein that is half-folded and half-disordered, what might one global number hide?

6 Open the models in PyMOL

The server returns **five** models per job (ranked by confidence). Download the ARL13B_apo results and unpack the zip. You will find five model coordinate files (*.cif) plus the confidence data.

Task. Open PyMOL and load all five apo models (drag the five .cif files into the PyMOL window, or use File>Open). Each loads as its own object.

The five files load with long names (based on what you called your job), so typing out alignment commands is fiddly. Instead, use the one built-in command that superposes every loaded object onto the first one, whatever they are named. Just type:

```
align to
```

Question. Where do the five models agree, and where do they fan apart? What do you think it means about a region when all five independent predictions land in the same place, versus when they scatter? Now compare the regions that scatter here with the regions that were yellow/orange in the browser: what is the relationship, and why would those two things go together?

7 Colour by confidence in PyMOL

For AlphaFold models the per-residue pLDDT is stored in the B-factor column, so you can reproduce the AlphaFold colouring. The model is already shown as cartoon, so just colour it (orange = low confidence, blue = high):

```
spectrum b, orange_yellow_cyan_blue, minimum=40, maximum=100
```

Optionally, make the cartoon thickness track confidence as well:

```
cartoon putty
```

Note. This colouring is specific to AlphaFold models, where the B-factor holds pLDDT (0–100, high = good). In an *experimental* structure the B-factor means something different (atomic displacement / disorder, high = worse), and you would colour it on a different scale, e.g. spectrum b, blue_white_red, minimum=20, maximum=50.

8 The switch regions: apo versus GTP-bound

Every small GTPase has **two** switch loops. **Switch I** carries a conserved threonine (here Thr52) that coordinates the Mg^{2+} and the nucleotide γ -phosphate; **switch II** follows the conserved DxxG motif (here from Asp71). These loops sense whether a nucleotide is bound. Select them in your apo model and put them on the confidence scale:

```
select switchI, resi 45-54
select switchII, resi 71-86
spectrum b, orange_yellow_cyan_blue, minimum=40, maximum=100
show sticks, (switchI or switchII) and not name C+N+O
```

Question. Find switch I (around residues 45–54) in the apo model. How confident is it compared with the rigid core of the domain? Switch I's job is to sense the nucleotide, so why might AlphaFold struggle to pin down its structure when the pocket is empty? What would you predict happens to it once GTP is bound?

Now bring in the second prediction.

Task. Download and unpack ARL13B_GTP and load a model into the *same* PyMOL session (do not open a new one). You now have both apo and holo loaded; switch the apo objects off with their checkboxes in the object panel on the right, so only the GTP model shows. Colour it by pLDDT with the same command as before. The GTP and Mg^{2+} are shown automatically; do not run as `cartoon`, which would hide them.

Question. Where did AlphaFold place the **GTP** and the Mg^{2+} : is it a sensible pocket, and does the ion sit where you would expect relative to the phosphates? (AlphaFold3 has predicted the ligand and the ion, not just the protein chain.) Look at which protein residues line the pocket: can you spot any recognisable GTPase sequence motifs among them?

Question. Recall your prediction from the apo section. Now look again at **switch I** in the GTP model: did its confidence change, and in which direction? Compare switch I with switch II: did they behave the same way? If not, what might explain the difference between the two loops?

Think about it. You have just watched one region change confidence simply by adding a ligand. So what does a low pLDDT actually tell you: that the prediction is *wrong*, or something else? Can you think of a case where low confidence is biologically *informative* rather than a failure?

Note. When everyone is done, we switch to a short summary presentation.

That is the core of Part 1. You have made two predictions and learned to read the pLDDT colouring, the spread of the five models, and how a flexible region can order up when a ligand binds.

Finished early? The optional section below introduces the **PAE**, the score we build on in Part 2, so it is a head start while others catch up.

Take-home: pLDDT is local; the spread of the five models is the uncertainty; low confidence (orange) means do not trust the local structure, including “helices”; a flexible region can order up when it binds a ligand or partner.

9 Optional extension: the PAE

Only if you finish the core exercise with time to spare.

Task. First work out for yourself what the **Predicted Aligned Error (PAE)** is and what it reports on (see the AlphaFold server’s own help). Then test it on your ARL13B_GTP model.

pLDDT is *local*: it says nothing about whether two parts of the model are placed correctly *relative to each other*. That is what the PAE adds. In the plot the colour at (x, y) is the expected error in residue x once the model is aligned on residue y : dark = low error (relative position trusted), light = high error.

Question. Open the PAE plot for ARL13B_GTP. Where is the error low and where is it high? Match those dark and light regions back to what you already know about this protein: which parts should be locked rigidly together, and which should be free to float? Does the plot agree with that expectation? The result also reports an **ipTM** for the ligand: what do you think that number is measuring, and would it make you trust where the GTP was placed?

Coming next (Part 2): predicting and validating a complex. We will predict a *protein–protein complex* and use the PAE and ipTM across the interface to judge whether the predicted interaction is real.